Design Of High Accuracy Detector For Mnist Handwritten Digit Recognition Based On Convolutional Neural Network

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***Abstract*—Tremendous strides have been made in machine learning, one of the remaining open challenges is to achieve real-time speed as well as to maintain high performance, A convolutional neural network is designed for MNIST handwritten digit recognition, where the first convolutional layer has 32 feature images, the second convolution layer has 64 feature images. Next is the full connection layer, where the first full connection layer has 2048 neurons, and the second_has 784 neurons. And the output layer has 10 neurons, which is corresponding to the 10 digit labels. In the experiment TensorFlow is adopted to setup network according to the above technology. When the training of network is completed after 500 epochs, the accuracy of the recognition is 96.1% for training set, and 95.7% for test set.**

Keywords- Deep learning; Convolutional Neural Network; Softmax Function; ReLU Function

I. INTRODUCTION

Pattern recognition is one of the fundamental problems in computer vision and artificial intelligence(AI). One of the major challenges of computer vision and AI in the next decade will be to devise methods that can automatically learn good features hierarchically from unlabeled and labeled data in an integrated fashion. The ultimate goal of artificial intelligence is to enable machines with the similar human beings' analytical ability, such as recognizing text, images, sounds and so on. With the great efforts of researchers many complex pattern recognition problems have been solved.

Deep Learning is a new research direction in the field of machine learning, which be introduced into machine learning to make artificial intelligence closer to the original goal -- replace the original manual processing, so as to improve theconvenience and reduce the rate of error due to manual operation greatly[1]. Convolutional neural network is a deep learning model or multilayer perceptron similar to artificial neural network. It is commonly used to analyze visual images. Yann Lecun, a famous computer scientist, is the founder of convolutional neural networks. He is the first person to achieve recognition of handwritten digit problems through adopting convolutional neural networks. It is widely believed that the beginning of deep learning was in 2006, in the large-scale visual image recognition competition ILSVRC in 2012, AlexNet proposed to return the neural network to the cusp of artificial intelligence, becoming a breakthrough model of convolutional neural networks. After showing the great advantages of convolutional neural networks, new models such as VGGNet and GoogLeNet have emerged. In this paper, according to the existing techniques we designed a convolutional neural network for handwritten digit recognition.

II. DESIGN OF DETECTOR HANDWRITTEN DIGIT BASED ON CONVOLUTIONAL NEURAL NETWORK

Convolutional neural networks are a kind of deep neural network. In 1998, Y. Lecun et al. combined the convolutional layer with the downsampling layer to establish a modern prototype LeNet of convolutional neural networks, which was later widely used in the recognition of handwritten characters. In the history of deep learning, convolutional neural networks play an important role in the improvement of computer vision research and practical performance[2-3]. The characteristics of local perception, weight sharing and sub-sampling in space or time make this kind of network have the invariance of translation and deformation.

A. Convolutional neural network common structure

There are two most obvious advantages of convolutional neural network, as are listed as following,

(1) Hidden feature extraction: convolutional neural network can extract the feature information from the original image directly, and needed preprocessing is very simple. Digit images can be used as the input of the network directly, and the feature extraction can be done at the same time of training.

(2) Local field and shared weights: The unique local field and parameter sharing mechanism of CNN make it highly invariant to digit translation, scaling, tilt, distortion, or other deformations.

At he same times, the pooling of convolutional neural network could reduce the resolution of the feature map and reduce the network size, which would reduce the sensitivity to translation, scaling and distortion.

Through experiments, it is found that the complexity of the network structure used has a great influence on the experimental results and training time[4-5]. In the

978-1-7281-4284-5/19/$31.00 ©2019 IEEE
DOI 10.1109/ICICTA49267.2019.00145

experiment, the model structure is divided into an input layer, a convolution layer, a pooling layer, a fully connected layer, and an output layer. The input layer is removed, and a total of 7 layers are included. The input is represented by x, usually a matrix or a picture, and the size is 28×28 pixels. The first and second layer convolution layers use a 5×5 size convolution kernel (convolutional filter) with a sliding step size of 1, and the first and third pooling layers are obtained by using a 4×4 size convolution map, The sliding stride size is 2. The 1st full connection layer have 2048 neurons, and the 2nd full connection has 784 neurons. The final output layer has 10 neurons, which uses the softmax algorithm for 10 classifications so that the output result of the pooled layer is directly connected to the fully connected layer[6-7].

Assume two-dimension discrete function(matrix) I(x, y) express original/*input* image, convolutional kernel function is $f(x, y)$, so their convolution is

$$Conv(m,n) = \mathrm{I}(x,y) \otimes F(x,y) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} I(m,n)F(x-m, y-n), \quad (1)$$

where $\otimes$ denote convolution operation, that is the convolutional kernel $f(x, y)$ slide in image I(x, y), so as to obtained a feature image.

In convolutional layer C_1 and C_2, there 32 and 64 feature maps are gotten respectively, through convolutional operation, the obtained features maps' height(n_h) and wide(n_w) is,

$$n\_h = n\_w = \frac{W-F}{S} + 1, \quad (2)$$

where W is the size of input images, F is the size of filter, and the S is the stride of slide. So the feature map in 1st convolutional layer is 24×24, the feature map in 2nd convolutional layer is 8×8.

As for the pooling layer, max-Pooling techniques is adopted, and the sliding stride size is 2. So similarly, in Fig.1, the feature maps in 1st pooling layer is 12×12, so the feature maps in 1st pooling layer is 4×4.

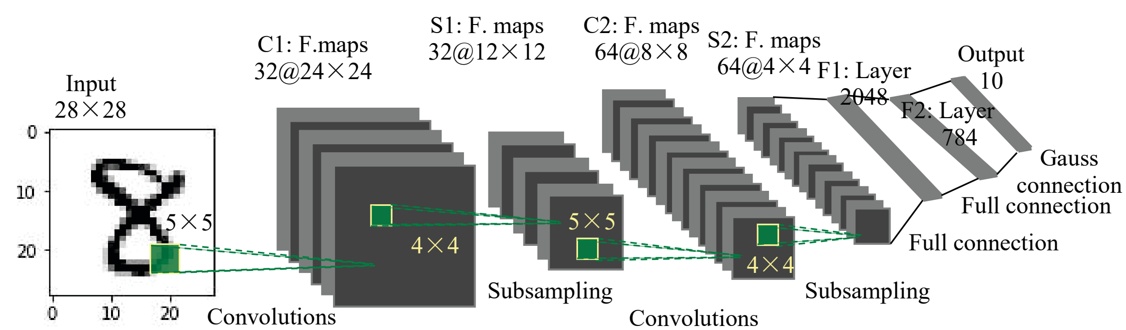

Fig.1 Framework of CNN for Handwritten Digit Recognition

The adopted activation function of the system is rectified linear unit (ReLU), that is,

$$f(x) = \max(0, \lambda x), \quad (3)$$

During the process of network training, the ReLU function has the advantage of fast calculation speed, and can effectively avoid gradient disappearance.

Softmax function is normalized exponential function. It is a generalization of sigmoid (bi-classification function) in multi-classification, i. e. that maps a length-p vector of real values to a length-10 vector of values

$$\sigma(z) = \frac{e^{z_j}}{\sum_{k=1}^{10} e^{z_j}}, \quad (4)$$

The program of handwritten digit recognition consists of three parts. The first part of the experiment is mainly used to establish the forward propagation process of the network structure and the determination of related parameters, and then define an inference (input train regularizer functions are easy to call when training and testing. The second part is the training part, through calling the defined defined function, setting the values of train and regularizer, then adding regularization, gradient descent algorithm and sliding average model to the training model, and finally setting the iteration mode to ensure every thousand iterations. After outputting the value of the loss function once, and the learning rate is 0.0007 while the system is trained. The third part is the test part. It still calls the inference function, sets the values of train and regularizer, directly calls the saved part of the training part, and then runs the test part. The program is developed with python 3.6 language in anaconda environment, the part codes is shown as following, and the corresponding detailed code is available at https://download.csdn.net/download/qq_39681854/11833516.

B. Tensorflow and tensorboard

TensorFlow is an open source software library for numerical computation using data flow graphs, which was originally developed by the Google Brain Team, a research institute of Google Machine Intelligence. The visual tensorflow is widely used in machine learning and deep neural network research.

Tensor Board is a set of Web program for checking and understanding TensorFlow running and graphics, which is one of the advantages of Google's TensorFlow against Facebook's Pytorch.

In this paper, through running the digit recognition program, a logs folder is obtained, According to an HTTP address is provided, then enter the prompted HTTP address in the browser, so a visual tensorboard was obtained, the visual tensorboard is shown in Fig. 2, which could be amplified for reader to check by reader-self.

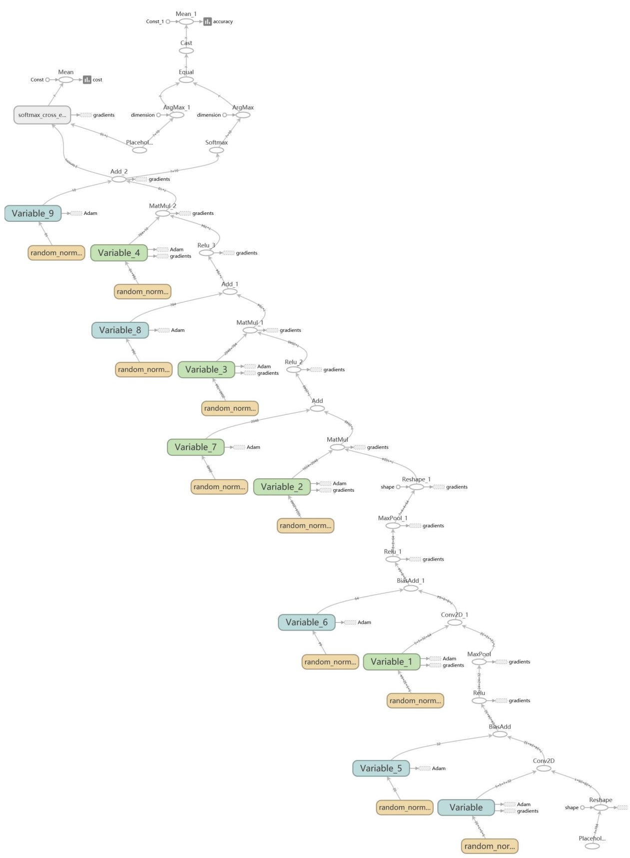

Fig.2 Visual Convolutional Neural Network

III. Experimental Introduction and Result Analysis

A. MNIST dataset

There are currently many public data sets that can be used, such as MNIST, ImageNet, Caltech-101/256, and so on. While the experiment is carried out, The MNIST database of handwritten digits is adopted in this experiment, which is the Mixed National Institute of Standards and Technology database. The MNIST contains training set and test set, and the training set consists of numbers written by 250 different people, 50% of whom are high school students, 50% from the census, and 50% from the census staff. The test set is also the same proportion of handwritten digital data. The training set has a set of 55,000 examples, and a test set of 10,000 examples. The MINIST digits have been size-normalized and centered in a fixed-size image, which sample description 28×28 pixels is a gray image, the length of the array 784 can be used to represent an image. The MNIST database can be available at http://yann.lecun.com/exdb/mnist/, the random 100 digits of the MNIST database are shown in Fig.3.

Fig.3 Partial digitals of the MNIST database

B. Experimental environment

While the high accuracy detector based on convolutional neural network for MNIST handwritten is trained, The experimental environment is 64bit operating system (Window10), Intel(R) Core(TM) i7-8705H CUP, clocked at 2.2Ghz, 8G internal memory, 128GB solid state hard disk, 1.81TB mechanical hard disk, and the adopted programming language is python3.6.

C. Experimental steps

There are three steps in the digit recognition experiment, the main steps are shown as following,

Step 1: Download the MNIST data set from web and enter the data set into the network.

Step 2: Set training part, determine the network structure and parameters, and select Add gradient descent algorithm updates the moving average model parameters, reduce the loss of function, and select L2 regularization method to prevent over-fitting.

Step 3: Adjust the networks' weights in the next step through the gradient negative feedback during back propagation.

While the training of neural network is completed, we can obtain the accuracy and loss of the system, which could be demonstrated with tensorboard. The accuracy of system is shown in Fig.4.

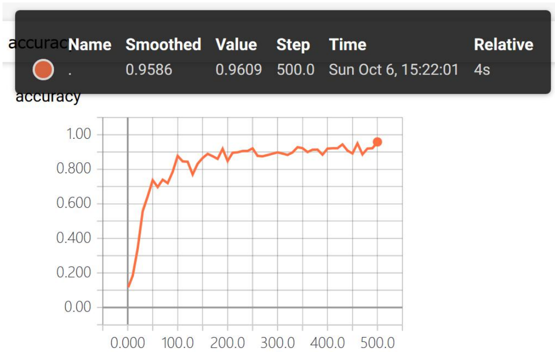

Fig.4 Accuracy of CNN for Training set

Can be seen from Fig.4, after 500 epochs, running time of the program is 4 seconds, the accuracy rate of the recognition has reached 96.09% for training set. And in the experiment, we know the recognition rate is 95.7% for test set. Through selected 100 digits from test set MNIST dataset randomly, there are 4 digit being recognized mistakenly, , for example, the actual digit of "9" is predicted as "4", which is shown in Fig. 5. We can draw that the designed network has high accuracy rate, which be applied in engineering practice.

Fig. 5 Results of recognition with the adopted techniques

And in order to compare with existing method, for example, the neural network is adopted. The designed neural network have 1 input layer, two hidden layer, and 1 output layer. The designed neural network, input layer have 784 neurons, first hidden layer have 1024 neurons, and the second layer have 512 neurons, and the output has 10 neurons, whose expect value are 10 labels, that is the 10 Arabic numeral. The program is with python 3.6, the loss function adopt "tf.reduce_sum(y_holder * tf.log(predict_y)", i. e. cross entropy. the code can be available at https://download.csdn.net/download/qq_39681854/11862045.

The training data is MNIST set also. When the neural network's training is completed, according to the experiment, we found that the accuracy of the recognition rate is 92.6% for training set, and 90.1% for test set.

IV. Conclusions

In this experiment, the research of the system involves many aspects of technology and theory, such as image recognition, image data preprocessing and so on. handwritten digit recognition algorithms in machine learning can be obtained relatively fairly. The recognition rate of convolution neural network is the higher. When the training of network is completed after 500 epochs, the accuracy of the recognition is 96.1% for training set, and is 95.7% for test set. Convolutional neural network is a powerful in-depth learning model, which is widely used and has excellent performance. With receptive field and value sharing, the convolutional neural network reduce the number of parameters needed to be trained. In addition, during the process of machine learning from the laboratory into the software, there will be many places needed to constantly be improved.

Acknowledgements

The work described in this paper is partially supported by National Natural Science Foundation of China under grant Nos. 51765007, 51675186, and the Guangxi Provincial Natural Science Foundation of China under grant No. 2016GXNSFAA380111. The authors would like to thank the reviewers for their constructive comments and substantial help that improved the presentation of the paper.

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